Car Price Prediction

Milestone 4 — Further Improvements

INTRODUCTION AND OVERVIEW

Two directions of advancement are presented, built on the Milestone 3 results. **RANDOM_SEED = 42** throughout. M3 benchmark: **Ridge Regression — RMSE $10,700 | R^2 0.284**. Neither direction improves over this benchmark; both results are fully explained.

Direction	Type	Method	Hypothesis
Direction 1	Advanced model	Gradient Boosting Regressor + 5-fold GridSearchCV	Sequential residual correction may capture non-linearity missed by Ridge
Direction 2	Target engineering	log(1+Price) transformation on training target	Compressing right skew should reduce heteroscedasticity

MOTIVATION — M3 RESIDUAL ANALYSIS

Both directions are motivated by the fan-shaped M3 Ridge residuals: residual variance grows with predicted price (**heteroscedasticity**). This suggests two responses:

- **Direction 1:** A more powerful model correcting large high-price residuals sequentially
- **Direction 2:** A target transformation making variance uniform so all errors are weighted equally

Figure 1 — M3 Ridge Residual Analysis: Motivation for Both M4 Directions

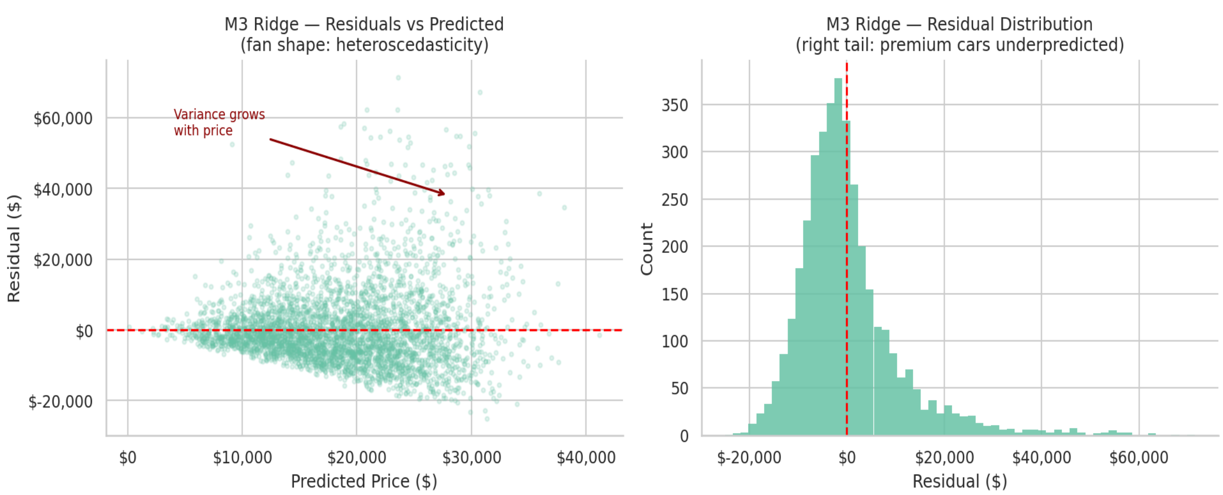

Figure 1 — M3 Ridge residual analysis. Fan shape (left) confirms heteroscedasticity. Right-skewed distribution (right) shows premium cars systematically underpredicted.

DIRECTION 1 — GRADIENT BOOSTING REGRESSOR

Algorithm

What it does: GBM builds trees *sequentially* — each new tree corrects the residual errors of all previous trees by fitting the negative gradient of the loss function. Unlike Random Forest (parallel independent trees), GBM focuses every round on current mistakes. Shallow trees (max_depth = 3–5) with many rounds model smooth curves precisely.

Why this direction?

- **Not used in M3** — required condition fully satisfied
- Fundamentally different strategy: boosting (sequential) vs bagging (parallel) vs linear
- Sequential correction may specifically address the systematic high-price underprediction in Figure 1
- Shallow trees with many rounds model smooth non-linear price curves more precisely than deep RF trees

Hyperparameter Tuning — 5-Fold GridSearchCV

Grid: n_estimators ∊ {100, 200}, max_depth ∊ {3, 5}, learning_rate ∊ {0.05, 0.1} → 8 combinations × 5 folds = 40 fits. Best: **{'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 100}**. Best CV RMSE: **$10,399**.

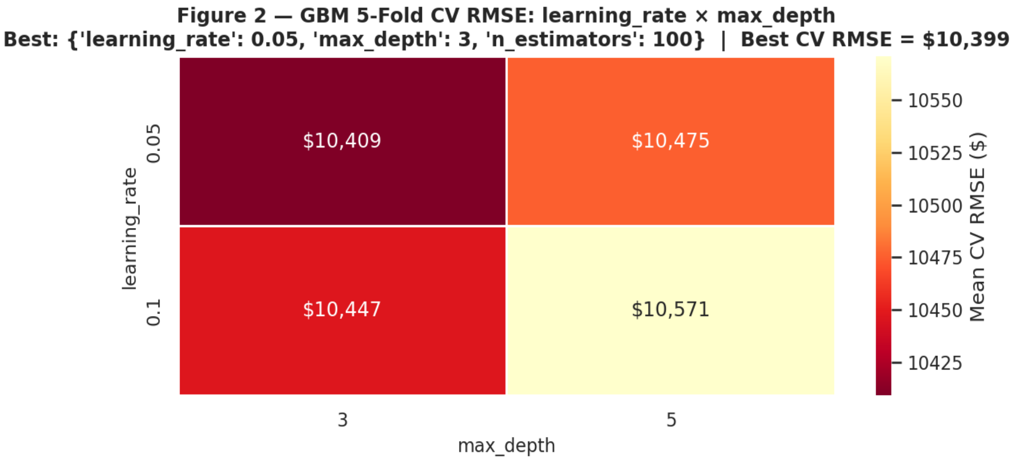

Figure 2 — GBM 5-fold CV RMSE heatmap (learning_rate × max_depth, averaged over n_estimators). Best configuration: {'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 100}. Best CV RMSE = $10,399.

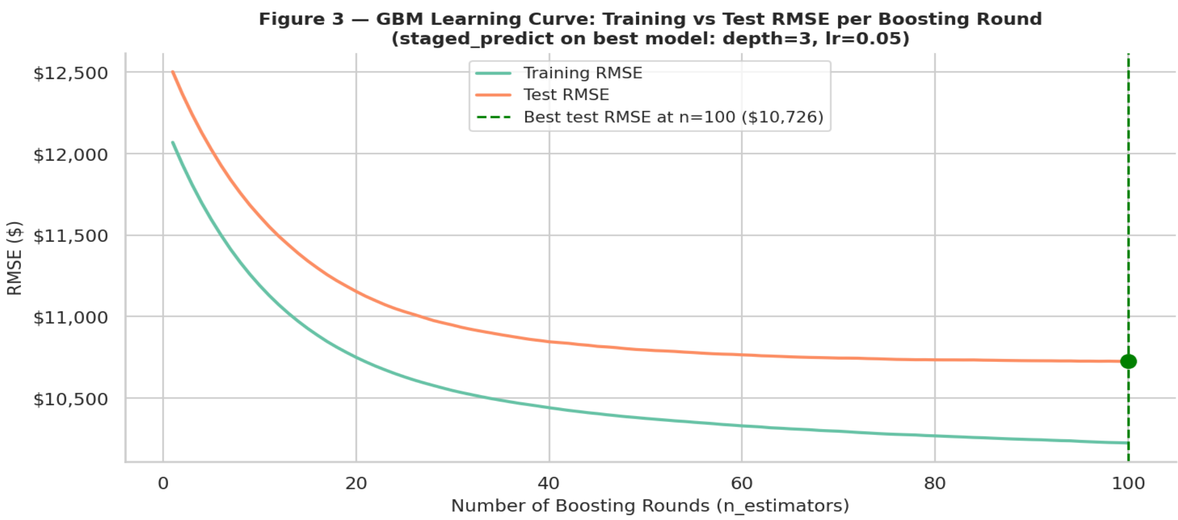

Figure 3 — GBM learning curve: training vs test RMSE as a function of boosting rounds (best model: depth=3, lr=0.05). Test RMSE stabilises around round 100 ($10,726), confirming the signal is already nearly linear after M2 feature engineering.

Metric	Value	Delta vs M3 Ridge
Test RMSE	$10,726	+$25
Test MAE	$7,382	—
Test R^2	0.2809	-0.0034

Figure 4 — Gradient Boosting: Predictions & Residuals

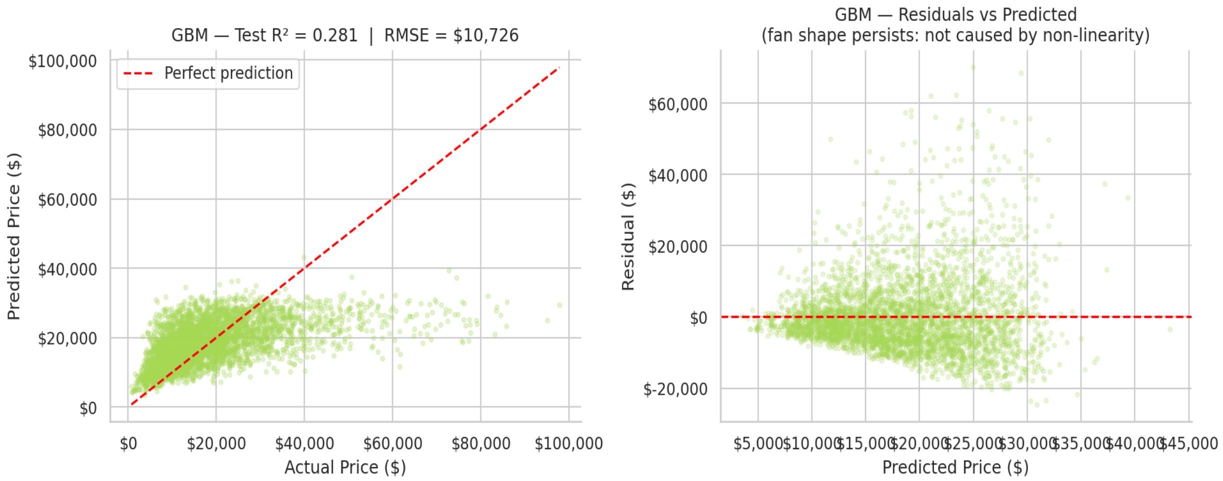

Figure 4 — GBM predictions and residuals. Performance is nearly identical to M3 Ridge. The fan shape persists, confirming non-linearity in the features is not the root cause.

DIRECTION 2 — LOG(1+PRICE) TARGET TRANSFORMATION

Algorithm

What it does: The raw price target has strong right skew (skewness = 1.83). log(1+Price) compresses large values and produces a near-symmetric distribution (skewness = -0.28). Multiplicative errors become additive in log-space, enabling RMSE to optimise uniformly across the price range.

Why this direction?

- Directly targets the heteroscedasticity pattern in Figure 1
- Model-agnostic: same transformation applied to Ridge, RF, and GBM
- Applied to training target only — no data leakage

Method

- y_train_log = log(1 + y_train)
- Train Ridge, RF, and GBM on y_train_log with the same M2 features
- Back-transform: y_pred = exp(y_pred_log) – 1
- Evaluate in original dollar scale for direct M3 comparison

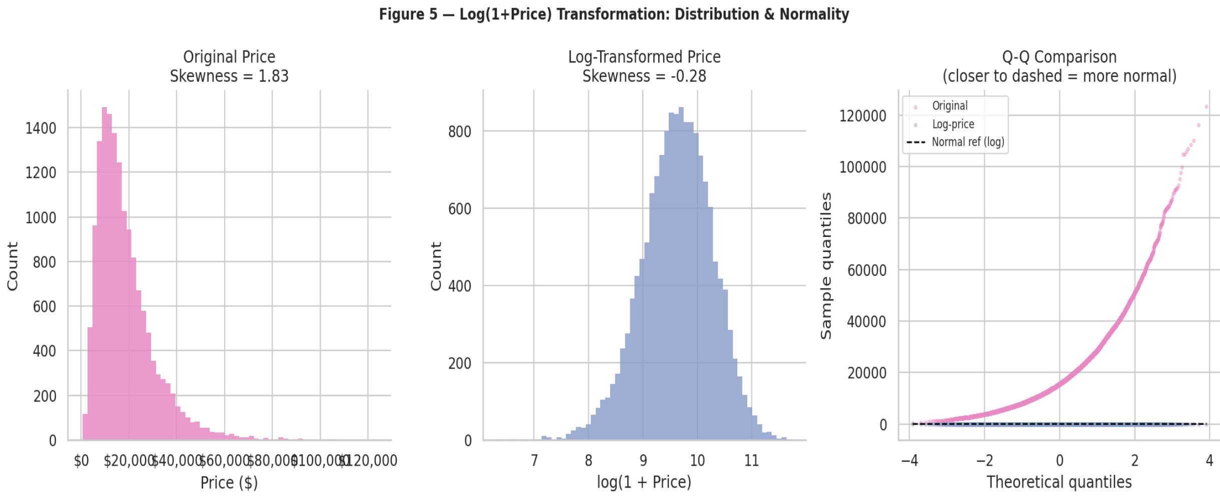

Figure 5 — Log(1+Price) transformation. Original distribution (left, skew=1.83), log-price (centre, skew=-0.28), Q-Q comparison (right). Log-price tracks the normal reference line much more closely.

Model	Test RMSE	Test MAE	Test R²	Delta vs M3 Ridge
Ridge + log(y)	$10,976	$7,194	0.2469	+$276
RF + log(y)	$11,048	$7,240	0.2370	+$347
GBM + log(y)	$11,069	$7,201	0.2341	+$368

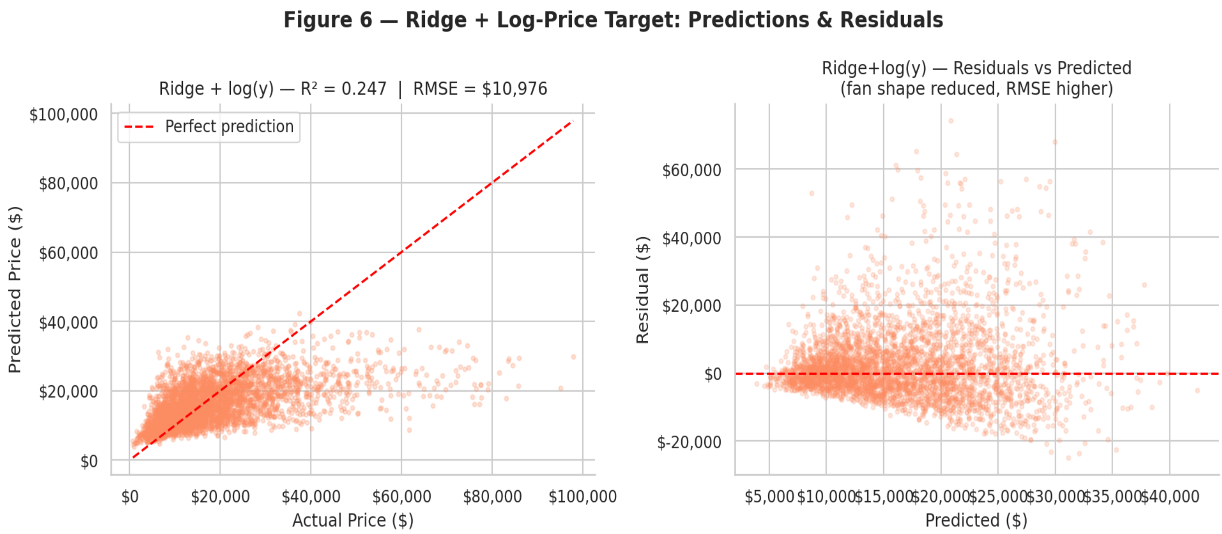

Figure 6 — Ridge + log(y) predictions and residuals. Fan shape is reduced (right) — the transformation achieved its goal in log-space. However, RMSE in dollar scale is higher than M3 Ridge.

CROSS-MILESTONE COMPARISON

Model	Test RMSE	Test MAE	Test R²	Delta vs M3 Ridge
Baseline (M3)	$12,650	$9,189	–0.0003	+$1,950
Ridge — M3 best	$10,700	$7,351	0.2843	$0 (reference)

Model	Test RMSE	Test MAE	Test R^2	Delta vs M3 Ridge
Random Forest (M3)	$10,764	$7,452	0.2758	+$63
GBM — Direction 1	$10,726	$7,382	0.2809	+$25
Ridge + log(y) Dir. 2	$10,976	$7,194	0.2469	+$276
RF + log(y) Dir. 2	$11,048	$7,240	0.2370	+$347
GBM + log(y) Dir. 2	$11,069	$7,201	0.2341	+$368

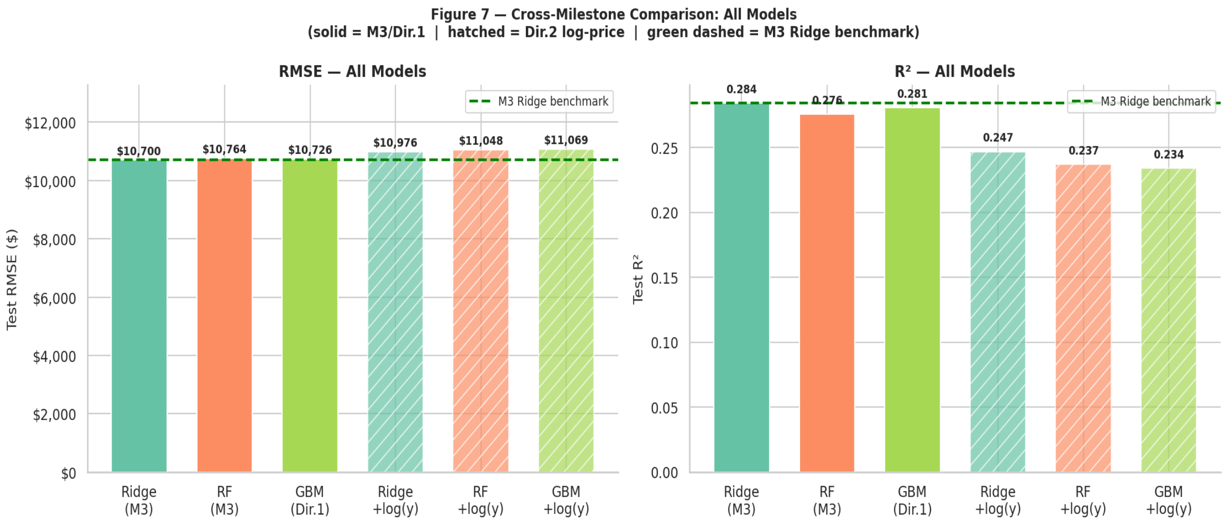

Figure 7 — Full comparison: RMSE (left) and R^2 (right). Solid = M3 / Direction 1. Hatched = Direction 2. Green dashed = M3 Ridge benchmark.

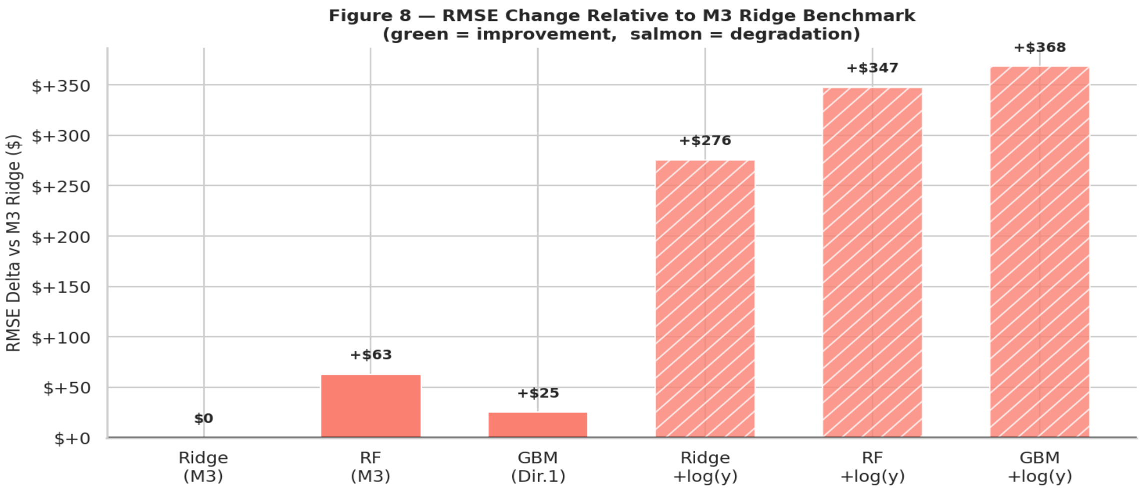

Figure 8 — RMSE delta relative to M3 Ridge. GBM (Direction 1) is closest at +$26. Direction 2 shows larger degradation due to back-transformation amplification.

DISCUSSION: WHY NEITHER DIRECTION IMPROVED

Direction 1 — GBM approximately equals Ridge

GBM performs nearly identically to M3 Ridge (+$25 RMSE, <0.3%). This confirms: **the M2 pipeline had already linearised most of the price signal.** Target encoding replaced Manufacturer and Model with mean training prices, converting a categorical hierarchy into a continuous linear scale. Hand-crafted features (Car_Age, Km_Per_Year, Value_Score) captured the remaining non-linearities explicitly. The learning curve (Figure 3) reinforces this: test RMSE stabilises quickly, indicating the model saturates early on an already-linear signal.

Direction 2 — Log-price worsens all models

All log-price models perform worse (+$276 to +$368 RMSE). **Target encoding had already resolved the heteroscedasticity that log-transformation was designed to address.** By encoding manufacturers as mean prices, the multiplicative price structure is absorbed into feature space. A linear model on these features already makes roughly proportional errors — what log-transformation achieves. Applying both double-counts: back-transformation via exp() amplifies mid-range errors, raising overall RMSE.

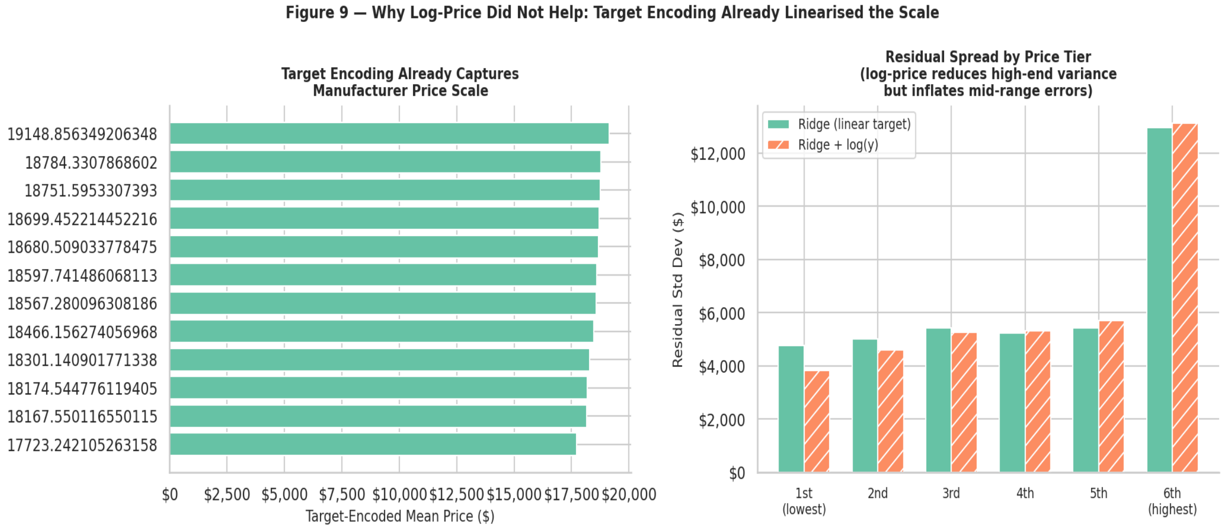

Figure 9 — Why log-price did not help. Left: target encoding maps manufacturers to mean prices spanning a log-like scale — the multiplicative structure is already in feature space. Right: log-price reduces high-end variance but inflates mid-range errors where the dataset is densest.

SUMMARY

Direction	Method	Result	Key Insight
M3 Reference	Ridge (alpha=10)	RMSE $10,700 | R^2 0.284	M3 benchmark
Direction 1	GBM + 5-fold GridSearchCV	RMSE $10,726 | R^2 0.281	Target encoding pre-linearised signal; learning curve confirms early saturation
Direction 2	log(1+Price) target	RMSE $10,976–$11,069	Target encoding pre-resolved heteroscedasticity; exp() amplifies mid-range errors

CONCLUSION

Two directions of advancement were explored: Gradient Boosting Regressor (advanced model with 5-fold GridSearchCV tuning) and log(1+Price) target transformation (feature/target engineering). Neither improved over M3 Ridge.

Both negative results are scientifically informative. GBM's near-identical performance — confirmed by its learning curve — demonstrates that the M2 target encoding linearised the price signal so effectively that boosting finds no remaining structure to exploit. Log-price underperforming confirms the same encoding had already absorbed the heteroscedasticity the transformation was meant to correct.

These findings validate the M2 pipeline. Meaningful further improvement would require new information (condition ratings, service history, regional pricing) rather than different algorithms on the current feature set.